

PEICHI HUANG

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Education

University of Michigan, Ann Arbor

Aug. 2024 – Present

Master of Data Science - GPA: 4.0/4.0

Michigan, United State

- Information Retrieval (A+), Machine Learning (A), Data Mining (A), Web Systems (A), Image Processing, Optimization of SIMPL, Parallel Programming with GPUs.

National Tsing Hua University

Sep. 2019 – Jan. 2024

B.S, Mathematics and Computer Science - Major-GPA: 3.84/4.0 (Academic Excellence Award)

Hsinchu City, Taiwan

Technical Skills

Programming Languages: C, C++, Python, Java, JavaScript, HTML, CSS, SQL, R, Kotlin

Developer Tools: Google Cloud Platform, Colaboratory, Git, PyTorch, TensorFlow, Scikit-Learn, Keras, Spark

Experience

BlockChain Security Corporation

Jun. 2025 – Aug. 2025

R&D Intern

Taipei City, Taiwan

- Constructed an OCR pipeline for identifying recovery seed phrases in real-world images by combining image processing, text recognition using PaddleOCR, and phrase validation using a Trie-based matching over the BIP-39 dictionary.
- Built a lightweight filtering system using Laplacian blur detection to improve efficiency and increase recall rate by 40%.

Information Technology Service Center, NYCU

Feb. 2024 – Aug. 2025

Research Assistant

Taipei City, Taiwan

- Evaluated the AI chatbot's performance on the center's website through NLP methods, and analyzed the chatbot usage.
- Performed exploratory data analysis (EDA) on post-graduate year medical student training outcome data and trained ML models such as regression and random forest with more than 0.85 accuracy to provide insight on future training.

Biomedica Corporation

Mar. 2024 – Aug. 2024

R&D AI Engineer Intern

New Taipei City, Taiwan

- Trained deep learning models with YOLO-v8 and YOLO-v9 frameworks to automate image segmentation, achieving an average MAP of over 0.90 and over 0.95 with image augmentation.
- Developed algorithms for image quality assessment, and disease diagnosis via segmentation with over 0.85 accuracy.

Institute of Information Science, Academia Sinica

Jul. 2022 – Jan. 2024

Software Engineer Intern

Taipei City, Taiwan

- Engineered a digital diary application using Android Studio, enabling the collection of clinical data from over 200 patients diagnosed with GERD while integrating an online questionnaire for comprehensive health reporting.

Publications

A Retrieval-based Chatbot for Customer Service Using Large Language Model

IEEE ICFTSS, 2024

Pei-Chi Huang (Best Presenter Award), Mei-Lien Pan

Kuala Lumpur, Malaysia

Effect of Training Content and Environment Factors on Resident Retention Rates

JCMA, 2025

Hsiao-Chin Shen, Pei-Chi Huang, Mei-Lien Pan, Ching-Weng Yang, et al.

Taipei, Taiwan, 2025

Projects

CUDA Optimization for CNN Convolution | C++, GPU Programming, CUDA

Fall 2024

- Optimized CUDA CNN convolution kernels using output parallelism, shared/constant memory, channel blocking, memory coalescing, and occupancy tuning on NVIDIA V100.
- Achieved $341\times$ speedup over a single-threaded CPU baseline (13.6 ms $\rightarrow$ 0.0398 ms) through profiling-driven kernel optimization and occupancy analysis with Nsight Compute.

SAIT, Study Abroad Information for Taiwanese Students | Vue, Git, Python, ML

Fall 2024

- Built a similarity-based admission analysis tool to recommend relevant study abroad cases using weighted embedding.

GUI-ASFormer, Detecting Keyframes in GUI Videos | Python, ML, DL

Winter 2025

- Improved ASFormer for GUI action segmentation with a two stream encoder, high-resolution embeddings, frame differencing, and multiscale temporal convolutions, boosting F1@50 by +7.1 and edit distance by +6.8.

Job Postings Searching System | Python, Information Retrieval

Fall 2024

- Implemented a system that used user profiles as queries to match relevant job postings, applying NLP techniques such as text classification, semantic similarity, and ranking models to achieve a 67% improvement in NDCG over baseline.